Skills Summary

Solving Trigonometric Equations on Restricted Intervals

Use this reference while completing your worksheet or when studying.

---

1. Linear trig equations

Method: isolate the trig function, find the *reference angle* from the positive value, then place one solution in each quadrant where that function has the required sign. Add the period, keep what lands in the interval.

Signs: sine + in QI, QII cosine + in QI, QIV tangent + in QI, QIII.

Periods: sin, cos: 2π tan: π (so a 2π-long interval holds *two* tangent solutions, not one).

Example: Solve $2\cos x + 1 = 0$ on $[0, 2\pi)$.

Step 1. One trig function, first power — isolate it and read the answer off the unit circle.

Step 2. $\cos x = -\frac{1}{2}$: reference angle $\frac{\pi}{3}$, and cosine is negative in QII and QIII.

Step 3. QII: $\pi - \frac{\pi}{3}$; QIII: $\pi + \frac{\pi}{3}$ $\Rightarrow$ $\boldsymbol{x = \frac{2\pi}{3},\ \frac{4\pi}{3}}$

Try it: Solve $2\sin x + \sqrt{2} = 0$ on $[0, 2\pi)$.

check: $\boldsymbol{x = \frac{5\pi}{4},\ \frac{7\pi}{4}}$

2. Rewrite with an identity, then factor

Rule: one angle, one function, then factor. An equation mixing $2x$ with x, or sin with cos, cannot be solved until a substitution makes the terms comparable.

$\sin 2x = 2\sin x\cos x$ $\cos 2x = 1 - 2\sin^2 x = 2\cos^2 x - 1$ $\sin^2 x + \cos^2 x = 1$

Never divide both sides by a factor containing the variable — dividing by $\cos x$ deletes every solution where $\cos x = 0$. Move everything to one side and factor instead.

Example: Solve $\sin 2x = \sin x$ on $[0, 2\pi)$.

Step 1. The two sides use different angles, so rewrite $\sin 2x$ to put everything at angle x, then factor rather than divide.

Step 2. $2\sin x\cos x - \sin x = 0 \;\Rightarrow\; \sin x\,(2\cos x - 1) = 0$

Step 3. $\sin x = 0$ gives $0, \pi$; $\cos x = \frac{1}{2}$ gives $\frac{\pi}{3}, \frac{5\pi}{3}$ $\Rightarrow$ $\boldsymbol{x = 0,\ \frac{\pi}{3},\ \pi,\ \frac{5\pi}{3}}$

Try it: Solve $\sin 2x = 2\cos x$ on $[0, 2\pi)$.

check: $\boldsymbol{x = \frac{\pi}{2},\ \frac{3\pi}{2}}$

3. Quadratic in a trig function

Method: if the equation has the shape $au^2 + bu + c = 0$ with $u = \sin x$ (or $\cos x$), factor it exactly as you would an ordinary quadratic, then solve each factor.

Then check each value of u: $\sin x$ and $\cos x$ only take values in $[-1, 1]$, so a factor giving $u = 2$ contributes *no* solutions — and $u = \pm 1$ contributes only *one* per revolution, not two.

Example: Solve $2\cos^2 x + \cos x - 1 = 0$ on $[0, 2\pi)$.
Step 1. Only $\cos x$ appears, to the second and first power — that is a quadratic, so factor in $\cos x$.
Step 2. $(2\cos x - 1)(\cos x + 1) = 0$
Step 3. $\cos x = \frac{1}{2}$ gives $\frac{\pi}{3}, \frac{5\pi}{3}$; $\cos x = -1$ gives the single angle π $\Rightarrow$ $\boldsymbol{x = \frac{\pi}{3},\ \pi,\ \frac{5\pi}{3}}$

Try it: Solve $2\sin^2 x - \sin x = 0$ on $[0, 2\pi)$.

check: $\boldsymbol{x = 0,\ \frac{\pi}{6},\ \frac{5\pi}{6},\ \pi}$

4. Periodic models

Model: $d(t) = \text{midline} + \text{amplitude} \cdot \sin\left(\frac{2\pi t}{\text{period}}\right)$.

Method: set $d(t)$ equal to the target, isolate the trig function, then substitute u for the whole inside angle. **Translate the interval too:** if $t \in [0, T)$ and the period is T, then $u \in [0, 2\pi)$ — one revolution, so the usual unit-circle count applies. Solve for u, convert each answer back to t, and state the units.

Example: Water depth is $d(t) = 5 + 3\sin\left(\frac{\pi t}{6}\right)$ metres, t hours after midnight. When is the depth 6.5 m on $[0, 12)$ hours?
Step 1. The target depth is a horizontal line across one period of the model, so isolate the sine and solve on the matching angle interval.
Step 2. $3\sin\left(\frac{\pi t}{6}\right) = 1.5 \Rightarrow \sin u = \frac{1}{2}$ where $u = \frac{\pi t}{6} \in [0, 2\pi)$.
Step 3. $u = \frac{\pi}{6}, \frac{5\pi}{6}$, and $t = \frac{6u}{\pi}$ $\Rightarrow$ $\boldsymbol{t = 1,\ 5}$ hours after midnight.

Try it: Same model. When is the depth 3.5 m on $[0, 12)$ hours?

check: $\boldsymbol{t = 7,\ 11}$ hours